

Example

- Consider the following equations of motion:

Dynamic equilibrium: $m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{r}_1 \\ \ddot{r}_2 \end{bmatrix} + k \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$

Free vibration: $\omega_1 = \sqrt{\frac{2k}{m}}, \quad \omega_2 = \sqrt{\frac{4k}{m}}$

$$\Phi_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

In modal space:

The generalised coordinates q , such that $r = \Phi q$

$$m\ddot{q} + kq = Q \quad \text{where} \quad \begin{cases} m = \Phi^T M \Phi \\ k = \Phi^T K \Phi \quad \text{and} \quad \Phi = [\Phi_1 \quad \Phi_2] \\ Q = \Phi^T R \end{cases}$$

- Assuming that $\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} R_0 \\ R_0 \end{bmatrix} \cos(\omega t)$ where R_0 is a constant, obtain the dynamic steady state response. 50%

- No loads are applied to DOF 2, and a constant load of magnitude R_0 is applied to DOF 1 for time $t > 0$. Calculate the response $r_1(t)$, assuming that the mass is at rest at its equilibrium position at time $t = 0$. 50%

$$1. \quad \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} R_0 \\ R_0 \end{bmatrix} \cos(\omega t) \quad \therefore \begin{bmatrix} \ddot{r}_1 \\ \ddot{r}_2 \end{bmatrix} = -\omega^2 \begin{bmatrix} R_0 \\ R_0 \end{bmatrix} \cos(\omega t) = -\omega^2 \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -\omega^2 m + 3k & k \\ k & -\omega^2 m + 3k \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} -\omega^2 m + 3k & -k \\ -k & -\omega^2 m + 3k \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

where $\Delta = (-\omega^2 m + 3k)^2 - k^2$

$$\therefore r_0 = \frac{1}{\Delta} \begin{bmatrix} -\omega^2 m + 3k & -k \\ -k & -\omega^2 m + 3k \end{bmatrix} \begin{bmatrix} R_0 \\ R_0 \end{bmatrix}$$

$$= \frac{1}{\Delta} \begin{bmatrix} -\omega^2 m + 2k \\ -\omega^2 m + 2k \end{bmatrix} R_0$$

$$2. \quad m = \Phi^T M \Phi = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = m \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = m \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$k = \Phi^T K \Phi = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} k \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = k \begin{bmatrix} 2 & -2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = k \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix}$$

$$Q = \Phi^T R = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} R_1 - R_2 \\ R_1 + R_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} R_0 \cos(\omega t)$$

$$m \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \ddot{q} + k \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix} q = \begin{bmatrix} R_1 - R_2 \\ R_1 + R_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} R_0 \cos(\omega t)$$

$$q = \int_0^t \frac{1}{\beta m} e^{-\alpha(t-\tau)} \sin(\beta(t-\tau)) R_0 d\tau$$

$$= \int_0^t \frac{1}{\omega \cancel{m}_{2m}} \sin(\omega(t-\tau)) R_0 d\tau$$

$$= \frac{R_0}{\omega \cancel{m}_{2m}} \left[-\frac{1}{\omega} \cos(\omega(t-\tau)) \right]_0^t$$

$$= \frac{R_0}{\omega \cancel{m}_{2m}} \left(\frac{1}{\omega} - \frac{1}{\omega} \cos(\omega t) \right) = \frac{R_0}{4k} (1 - \cos(\omega_1 t)) \quad \text{for DoF 1, } \omega_1 = \sqrt{\frac{2m}{R}}$$

$$\text{or } = \frac{R_0}{8k} (1 - \cos(\omega_2 t)) \quad \text{for DoF 2, } \omega_2 = \sqrt{\frac{4m}{R}}$$

$$\therefore q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \frac{R_0}{4k} \begin{bmatrix} 1 - \cos(\omega_1 t) \\ 2 - 2\cos(\omega_2 t) \end{bmatrix}$$

$$\therefore r = \Phi q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} q = \frac{R_0}{4k} \begin{bmatrix} 3 - 2\cos(\omega_2 t) - \cos(\omega_1 t) \\ 1 + \cos(\omega_1 t) - 2\cos(\omega_2 t) \end{bmatrix}$$